The OECD-WTO Balanced Trade in Services Database (BaTIS)

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Acknowledgements

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Executive summary

Complete, consistent and balanced bilateral trade in services statistics are vital for the empirical analysis of international trade as well as for policy-making and trade negotiations. Unfortunately, such data are not readily available. This paper presents the work of OECD and WTO to build the Balanced Trade in Services (BaTIS) dataset, a complete and consistent trade in services matrix to serve as input for the compilation of the TiVA Inter-Country Input-Output Tables and as a tool for the analysis of trade patterns in general. This edition of BaTIS provides annual data for 2005-23, covering 202 economies, broken down by total services and 26 EBOPS 2010 service categories. This paper accompanies the dataset and describes its compilation methodology in detail, including the collection and cleaning of the reported data, the different methodologies used to estimate missing information, and the final balancing of the export and import flows.

1. Introduction

1. Demand for reliable bilateral services trade data has been growing in the past decade. International trade in services is playing an increasingly important role in international trade. Indeed, in 2023, services trade accounted for approximately one quarter of global trade, up from one fifth a decade earlier. Furthermore, according to the WTO RTA (Regional Trade Agreements) database, 373 RTAs were currently in force as of December 2024, of which an increasing number include a services chapter. Services trade is also closely related to digital trade which is now very high on the policy agenda and for which a statistical framework has now been internationally agreed. (IMF et al., 2023).

2. However, the availability of high quality, detailed information on international trade in services remains inadequate for the needs of economic analysis and policy-making. To address this information gap, the OECD and WTO have developed the OECD-WTO Balanced Trade in Services (BaTIS), a global dataset of coherent bilateral trade in services statistics by service category. Its methodology leverages all official statistics available at national level and supplements them with estimations and adjustments to provide users with a complete matrix of exports and imports, covering virtually all economies in the world. Subsequently, the asymmetries between reported and mirror flows are reconciled by calculating a symmetry-index weighted average between the two, following a similar approach to the one developed for merchandise trade statistics (Cassimon et al., forthcoming).

3. The ultimate goal of the work is to develop and maintain a dataset that can be considered the international benchmark for trade in services statistics, and which will be constantly improved as new official data become available. While the BaTIS dataset can serve as a stand-alone tool for economic analysis and policy-making, it also provides an essential input to the OECD Trade in Value Added (TiVA) initiative, as a balanced view of international trade is an essential feature of any inter-country Input-Output analysis.

4. Since the publication of the first BaTIS dataset in 2017, several advancements have been made in the compilation of trade in services statistics. Firstly, more and more economies have been working to improve their data compilation methods and practices, bringing them closer to the international guidelines. At the same time, increased emphasis has been placed on another dimension of international trade in services, namely the mode of supply. The relevance of this dimension in trade negotiations motivated the WTO to create a global dataset on Trade in Services by Mode of Supply (TISMOS, see Wettstein et al., 2019).

5. The fourth edition of BaTIS covers the period 2005-23, with more detailed EBOPS 2010 services categories (14 two-digit EBOPS 2010 sub-categories, in addition to the 12 main EBOPS 2010 services categories previously published).[1]

6. This technical paper describes the methodology of the BaTIS dataset together with key insights from the latest edition.

- While the difference between reported and balanced values is minimal at world level, it can be quite substantial for individual traders. In particular, it appears that some countries systematically under- or over-report their trade in services compared with what their partners declare, resulting in significant differences between the sum of the balanced trade figures and the original reported totals.

- The aggregate absolute differences between the reported data and the final balanced estimates are around 1% to 4% at world level. Those differences are generally much larger, however, for the individual service categories. Travel is the most aligned category at world level.
- High asymmetries are also observable for Telecommunication, computer and information services and Government goods and services n.i.e, and Financial services.
- According to the BaTIS dataset, world trade in services is estimated at around 7.6 trillion USD in 2023, with Transport, Travel and Other business services, accounting for more than 60% of world services trade. Over the 2005-23 period, services trade has increased by an average annual rate of 6%, overcoming significant troughs during the Great recession (2009) and the Covid-19 pandemic.
- Digitalisation has had a considerable impact on trade in services, as technological advances allow the remote delivery of many types of services. Hence, over the 2005-23 period, at the world level, digitally deliverable services grew at an average annual rate of 7.4%, higher than the 4.7% recorded for non-digitally deliverable services.

7. The remainder of the paper is structured as follows. Section 2. describes collection and cleaning of reported information (Step A). Section 3. describes the estimation of non-reported bilateral flows (Step B). Section 4. presents the balancing of import and export flows (Step C). Key results are presented and discussed in Section 5. . Finally, Section 6. concludes and outlines areas for future work and research.

2. Step A: Data collection and cleaning

2.1. Trade in services with partner world

8. As BaTIS is built following a top-down approach, the official statistics on international trade in services with partner world constitute the basic building blocks and provide the boundaries for all the subsequent estimations. The WTO-UNCTAD trade in services dataset,[2] arguably the most comprehensive set of information publicly available, has been used as a starting point for BaTIS. It covers exports and imports of services to and from the world for 202 reporting economies from 2005 to 2023, broken down by service category according to the availability of the source data. For the purposes of BaTIS, the 12 main service categories as well as 14 sub-services categories from EBOPS 2010 have been selected for publication (Table 1).[3]

Table 1. EBOPS 2010 categories classification: Code names and hierarchy

Code	EBOPS 2010 category description	Type	Derivation
S	Total services	standard	
SA	Manufacturing services on physical inputs owned by others	standard	
SB	Maintenance and repair services n.i.e.	standard	
SC	Transport	standard	
SC1	Sea transport	standard	
SC2	Air transport	standard	
SC3	Other modes of transport	standard	
SC4	Postal and courier services	standard	
SD	Travel	standard	
SDA	Travel; Business	standard	
SDB	Travel; Personal	standard	
SE	Construction	standard	
SF	Insurance and pension services	standard	
SG	Financial services	standard	
SH	Charges for the use of intellectual property n.i.e.	standard	
SI	Telecommunications, computer, and information services	standard	
SI1	Telecommunications services	standard	
SI2	Computer services	standard	
SI3	Information services	standard	
SJ	Other business services	standard	
SJ1	Research and development services	standard	
SJ2	Professional and management consulting services	standard	
SJ3	Technical, trade-related, and other business services	standard	
SK	Personal, cultural, and recreational services	standard	
SK1	Audiovisual and related services	standard	

Code	EBOPS 2010 category description	Type	Derivation
SK2	Personal, cultural, and recreational services other than audiovisual and related services	standard	
SL	Government goods and services n.i.e.	standard	
	Memorandum items:		
SOX	Commercial services	derived	S - SL
SOX1	Other commercial services	derived	SE + SF + SG + SH + SI + SJ + SK
SPX1	Other services	derived	SE + SF + SG + SH + SI + SJ + SK + SL
SPX4	Goods-related services	derived	SA + SB

Source: UNSD, Extended Balance of Payments Services Classification 2010 (EBOPS 2010).

9. The WTO-UNCTAD dataset combines reported data from Eurostat, the OECD, the IMF Balance of Payments Statistics, as well as national sources. In addition to the reported information, the dataset already includes a number of adjustments made to ensure cross-country comparability and, where possible, estimations to complement reported data (see Wettstein et al., 2019).

Table 2. Summary of reported, estimated and corrected exports and imports in the WTO-UNCTAD dataset (partner world), 2005-23

Percentage

	Exports		Imports	
	% Count	% Value	% Count	% Value
Reported	90.7	94.5	90.0	92.5
Estimated / adjusted / derived	9.3	5.5	10.0	7.5
Total	100	100	100	100

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

10. As shown in Table 2, the reported information in the WTO-UNCTAD dataset covers more than 90% of the value of world trade in services. For a number of reporters sourced from Eurostat, the time series start after 2005 (mostly in 2010) and the time series were completed with estimates based on information published at national level. Other adjustments include, for example, the conversion to a free on board (f.o.b.) valuation, the conversion to the BPM6 presentation, or corrections for obvious misclassifications, as well as derivations from existing items, which exploit the parent-child relationships in EBOPS. Finally, completely missing time series (at the level of service items) were estimated using average shares for clusters of economies showing similar characteristics.

11. Two additional adjustments were made to the WTO and UNCTAD dataset:

- Correction of negative insurance flows. When events, such as natural catastrophes, require large payments of claims, and exceed the value of premiums in the accounting period, the reported trade in insurance services can be negative.[4] As the interpretation (and treatment) of those negative flows is not straightforward, the negative values were replaced with zeros and total services adjusted accordingly. A similar approach was used to treat negative values and solve internal consistency problems for the bilateral dataset (see Section 2.2).

- Treatment of unallocated trade (by service category). Whenever the internal consistency between the reported parent service and the sum of its underlying children did not hold, some adjustments were needed. In the case of positive residual trade, the difference was allocated proportionally across the underlying EBOPS parent category. When negative residual trade existed, total services and the other parent categories were adjusted in line with the sum of the reported children service categories.

2.2. Bilateral trade in services

12. As a second step, all available bilateral trade in services information, i.e. all information on trade in services by partner country, was collected from the following sources: the OECD International Trade in Services statistics; Eurostat International Trade in Services statistics; Eurostat Quarterly Balance of Payments statistics; UN Comtrade; official national sources; and complementary national data.[5] Overall, for BaTIS, only 67 economies report full or at least partial bilateral information.

13. Table 3 gives an overview of the proportion of total world trade in services that is bilaterally specified, by year, flow and EBOPS category. As the table shows, there is an improvement in coverage over time, with more than 65% of total world services trade being bilaterally specified towards the end of the period. Exports show a better coverage than imports, as advanced economies with more developed statistical systems tend to be larger service exporters (also, exports of services are generally considered to be better reported than imports). The coverage is however significantly lower for some main service categories such as construction, insurance and pension services, or personal, cultural, and recreational services. For most two-digit EBOPS categories, coverage is lower compared to that of their one-digit parent categories. In cases where trade is concentrated among mostly advanced economies, the coverage of bilateral information is relatively high (like for charges for the use of intellectual property n.i.e).

14. Some economies report bilateral trade flows with partners outside the list of 202 economies covered by BaTIS. This happens most often for some EU members, which publish a large list of trading partners for total services. Those additional partner economies (which include small territories such as American Virgin Islands, Marshall Islands, Isle of Man, etc.) are summed up into the 'Rest of the World' (ROW) aggregate. Occasionally, this aggregate may represent between 3% and 5% of total trade for some reporters in specific years, but on average it accounts for less than 1% at world level.

15. Further information about the availability of bilateral trade in services statistics and the composition of the aggregate ROW is presented in Annex A. The maximum number of partner countries per reporter and sector is shown in Annex B.

Table 3. Bilaterally specified trade in services, by category

Percentage of total value

		Exports						Imports					
Service category	Code	2005	2010	2015	2020	2022	2023	2005	2010	2015	2020	2022	2023
Total services	S	36	60	72	71	67	52	34	55	65	65	63	48
Manufacturing services on physical inputs owned by others	SA	19	43	73	79	55	54	66	70	71	74	65	44
Maintenance and repair services n.i.e.	SB	28	67	68	59	56	43	23	63	72	68	63	50
Transport	SC	35	55	65	62	56	41	30	44	52	48	47	35
Sea transport	SC1	4	33	34	38	33	7	8	21	23	24	26	8
Air transport	SC2	16	36	40	42	40	21	14	31	35	35	38	24
Other modes of transport	SC3	10	36	61	65	63	2	5	37	53	59	56	2
Postal and courier services	SC4	7	25	30	35	36	6	6	33	26	38	26	4
Travel	SD	30	52	62	59	59	49	37	50	60	43	50	38
Travel; Business	SDA	21	35	47	48	52	25	14	30	31	36	40	13
Travel; Personal	SDB	13	26	37	45	42	18	11	29	30	34	43	20
Construction	SE	11	25	41	37	31	27	9	26	40	45	32	28
Insurance and pension services	SF	40	61	62	62	60	53	31	48	45	49	44	42
Financial services	SG	28	64	68	70	69	52	21	62	74	75	74	50
Charges for the use of intellectual property n.i.e.	SH	49	68	79	80	77	69	38	55	59	66	66	60
Telecommunications, computer, and information services	SI	20	53	56	57	53	43	24	61	63	62	61	49
Telecommunications services	SI1	2	29	24	35	33	14	1	31	30	38	38	11
Computer services	SI2	2	24	33	37	32	8	2	45	36	41	42	12
Information services	SI3	4	38	48	55	57	25	5	25	41	44	50	12
Other business services	SJ	30	57	66	65	63	53	28	55	65	63	68	52
Research and development services	SJ1	2	37	61	63	66	30	1	30	61	45	67	23
Professional and management consulting services	SJ2	3	39	46	56	51	26	3	40	55	61	62	21
Technical, trade-related, and other business services	SJ3	2	27	39	51	50	19	1	23	36	48	46	16
Personal, cultural, and recreational services	SK	15	51	53	61	71	57	28	40	41	46	46	40
Audiovisual and related services	SK1	7	59	56	59	63	44	7	36	34	31	40	23
Personal, cultural, and recreational services other than audiovisual and related services	SK2	0.3	12	15	25	53	11	1	10	15	19	24	4
Government goods and services n.i.e.	SL	19	25	30	33	29	18	24	27	25	23	34	23

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

16. Before proceeding to the subsequent estimations, the following cleaning steps were carried out on the collected bilateral data:

- Treatment of negative values. Negative trade flows are sometimes reported by the primary sources (most often for categories such as manufacturing services on physical inputs owned by others, maintenance and repair services n.i.e, insurance and pension services, and in some rare cases also for total services). As negative trade flows have no economic rationale, those values were corrected as follows:
  - For manufacturing services on physical inputs owned by others and maintenance and repair services n.i.e, the negative values were replaced by zeros.
  - For insurance and pension services, the negative values were replaced with zeros and the total services were adjusted accordingly for the reporter of such negative flows. If mirror flows existed, the corresponding amounts were added to the mirror flow and total services were recalculated in order to maintain the reported trade balance.[6]
  - For all other service categories, negative values were removed and replaced with estimates as described in step B.
- Treatment of unallocated trade (by service category). When positive residual trade existed, the difference was allocated proportionally across the EBOPS subcomponents. When negative residual trade existed, total services was replaced with the sum of the reported items.

17. Finally, to ensure hierarchical consistency, firstly between total services and the 12 main one-digit services items, then between 5 main one-digit services and their two-digit children services, the following rules were applied:

- If all children EBOPS categories were reported and their sum was not equal to their reported parent service, the parent was replaced by the sum of its children items.
- If fewer EBOPS children categories were reported and the sum of the children items was greater than the reported parent service, the parent was replaced by the sum of its individual children items.
- If the parent service category was positive and all children categories were reported as zero, all children service categories were removed and replaced by estimates as described in step B.

3. Step B: Estimating missing trade in services

18. After Step A, missing bilateral observations were estimated in order to obtain a complete trade matrix.

19. To perform this estimation, once again, a 'top-down' approach was used: a full matrix of bilateral flows was first built at the level of total services, then broken down by the main one-digit service categories, and finally broken down the more detailed two-digit service categories. This approach reflects the assumption that the greatest amount and best quality data is available at the highest levels of aggregation and ensures full consistency of the lower-level estimates. More specifically, the estimation work was organised in three sub-steps:

- Step B.1: Estimating partner country breakdowns if some partner data are reported.
- Step B.2: Estimating partner country breakdowns if no official data are available.
- Step B.3: Ensuring the consistency of the dataset.

3.1. Step B.1: Estimating partner country breakdowns if some partner data are reported

20. Step B.1 involves the filling of all data gaps in existing time series. A time series exists if at least one data point (year) is reported for a given combination of reporter-partner-flow-category. When bilateral trade in services statistics are reported, in fact, they rarely cover the entire time span considered (2005- 23) and the complete list of service categories. This estimation step therefore uses derivations, backcasting, nowcasting and interpolation techniques to fill in the missing information. Gaps in total trade in services are always filled in first, and subsequent estimations for the sub-items are then rescaled to maintain internal consistency.

Simple derivations from reported items

21. A number of additional data points in the bilateral dataset were derived exploiting the parent-child relationships in the EBOPS hierarchy:

- Firstly, if, for a country pair, total bilateral trade (S) is zero, then all main one-digit service categories are assumed to be zero for that particular pair/year.
- Then, if, for a country pair, any one-digit service category is zero, then all its two-digit children categories were assumed to be zero for that particular pair/year.
- If, for a specific category and year, a country reports zero trade with the world, then all bilateral values were assumed to be zero.

Backcasting, nowcasting and interpolation

22. This step exploits the information contained within the time series. If a time series was reported for at least one year, and contained at least one missing value, either in the middle, at the beginning or at the end of the time series, the missing data point was interpolated, backcasted or nowcasted, respectively.

23. Starting with total services, three-year moving averages of the partner's share in total world trade were computed. Those shares were then applied to the world values to get an estimate of bilateral trade. The same approach was also applied to the bilateral breakdowns of individual EBOPS categories, which were subsequently benchmarked to the total services values.

24. Table 4 shows an example of application of these techniques to the bilateral trade between France and Spain.

Table 4. Exports of services from France to Spain, by year and EBOPS category

Millions of USD

	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
S	6 365	6 942	8 186	9 230	7 991	8 012	10 457	10 262	10 759	11 482	10 164	11 190	12 284	15 296	18 040	11 044	14 256	16 189	18 130
SA	28	35	37	32	61	25	99	387	184	177	259	200	213	388	466	381	947	1 569	1 723
SB	45	50	59	62	80	80	161	141	113	162	98	148	158	242	230	238	294	323	378
SC	990	1 082	1 302	1 365	1 092	1 479	1 942	1 883	2 372	1 957	1 469	1 465	1 556	1 846	1 775	1 658	2 199	2 582	2 186
SC1	73	107	149	152	101	175	188	205	304	493	346	353	434	503	486	480	890	1 171	1 138
SC2	141	156	180	202	156	170	278	175	103	110	123	161	150	196	237	55	76	117	103
SC3	742	784	942	990	815	1 117	1 450	1 481	1 956	1 328	976	911	910	1 098	1 007	1 041	1 136	1 208	881
SC4	34	35	31	21	20	17	26	22	10	25	24	39	61	49	46	82	97	85	63
SD	2 652	2 839	3 334	4 046	3 252	2 645	3 111	3 035	2 802	3 161	3 042	2 948	3 631	4 898	5 053	2 675	3 526	4 472	4 998
SDA	399	428	510	639	479	424	596	400	505	454	622	683	451	731	774	278	588	301	318
SDB	2253	2 412	2 824	3 407	2 773	2 221	2 514	2 636	2 297	2 707	2 420	2 265	3 181	4 166	4 279	2 397	2 939	4 170	4 680
SE	61	64	77	83	70	83	2	3	6	7	5	1	3	7	0	2	0	1	0
SF	116	132	132	191	233	92	273	116	201	165	102	256	386	481	571	416	674	484	842
SG	79	90	125	83	109	209	231	382	469	355	323	249	242	224	237	364	519	473	731
SH	249	268	298	401	284	260	353	207	288	276	416	691	629	647	840	420	335	328	720
SI	371	413	498	492	541	538	768	699	868	665	753	1 159	1 186	1 484	1 874	1 396	1 888	1 877	1 823
SI1	169	188	227	220	241	245	363	305	337	206	225	329	263	367	401	208	263	246	229
SI2	191	213	257	258	284	277	384	364	482	431	487	776	879	1 040	1 372	1 094	1 471	1 466	1 431
SI3	11	12	14	14	16	16	22	30	49	28	41	54	44	77	101	95	154	166	162
SJ	1 728	1 890	2 253	2 442	2 173	2 572	3 335	3 361	3 395	4 488	3 641	3 953	4 183	4 759	6 716	3 442	3 727	3 923	4 544
SJ1	539	589	673	707	641	803	857	982	1 247	1 826	1 526	1 114	1 331	1 344	1 591	734	718	803	963
SJ2	480	509	593	630	569	666	738	662	785	1 124	703	817	1 154	1 031	1 301	1 337	1 493	1 496	1 697
SJ3	709	792	987	1 104	963	1 103	1 739	1 717	1 364	1 537	1 412	2 022	1 698	2 384	3 824	1 371	1 516	1 623	1 884
SK	23	24	29	34	26	30	182	48	60	70	54	118	97	306	277	52	147	158	186
SK1	22	24	28	34	26	30	180	48	59	69	53	110	93	287	269	51	144	154	180
SK2	0	0	0	0	0	0	1	1	1	1	1	8	4	19	9	1	3	4	6
SL	24	54	43	0	69	0	0	0	0	0	0	0	0	15	0	0	0	0	0

Note: Shaded cells represent backcasted and interpolated figures.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

3.2. Step B.2: Estimating partner country breakdowns if no official data are available

25. At the end of step B.1, 75% of the complete matrix in terms of data points remains to be filled, covering all the cases where no reported data are available for a given reporter-partner-flow-category combination. Those gaps are filled using a set of gravity models. Intuitively, gravity models relate trade flows between two countries positively to their economic size (proxied by the reporter and partner's GDP) and negatively to the trade costs (e.g. their geographical distance).

26. The gravity models were defined as parsimoniously as possible, and by explicit design, do not include any policy variables. This choice facilitates the subsequent use of the estimated values for analytical purposes, as it limits the problem of circular causality whereby the data that are used to test particular hypotheses (e.g. of the impact of certain policy measures on trade flows) are developed exactly based on that premise.

Gravity model specification

27. All regression models used in BaTIS share the following generic specification:

$$X_{ijt} = \exp\left(\beta_0 + \beta_1 size_{it} + \beta_2 size_{jt} + \beta_3\ trade\ costs_{ij} + \beta_4\ other\ predictors_{ijt}\right) * \varepsilon_{ijt}$$
$$M_{ijt} = \exp\left(\beta_0 + \beta_1 size_{it} + \beta_2 size_{jt} + \beta_3\ trade\ costs_{ij} + \beta_4\ other\ predictors_{ijt}\right) * \varepsilon_{ijt}$$

28. where Xijt (Mijt) represents the service exports (imports) of country i to (from) country j in year t, which are deemed to depend on:

- the size of the two trading partners, proxied by their nominal GDPs (sourced from the IMF World Economic Outlook database, and complemented by World Bank WDI, UN Data and WTO estimates)
- a set of trade cost variables, including the bilateral distance, contiguity, common language, and the presence, at any point in time, of a colonial relationship (all sourced from the CEPII GeoDist database and complemented with manual imputations for countries not covered by CEPII)[7]
- additional predictors, including bilateral (total) merchandise exports and imports (sourced from the UNCTAD Merchandise Trade Matrix[8]), the number of total tourist arrivals (exports) and departures (imports) in/from a country (sourced from UNWTO[9]), exports and imports of services to the world as a whole, a linear time trend, the reporter's GDP per capita and partner fixed effects.

29. Bilateral merchandise trade proved to be a very important predictor of bilateral service flows as trade in goods and services are generally highly correlated (especially for some service categories such as transport). Tourist arrivals in a country also performed relatively well as predictor, given the importance of travel in total trade in services (around a fifth of total services at world level but significantly higher for some economies).

30. Services trade with the world is used to predict missing flows at the level of total services and individual service categories. In the previous edition of BaTIS, a cross validation exercise demonstrated that the inclusion of this predictor, despite the endogeneity with the response variable, significantly improves the predictive power of the models and was therefore used to predict all missing bilateral flows (Annex F). The sample used for the cross-validation exercise contained information available for all specifications. The algorithm took the following steps:

- The sample was randomly split into two datasets: the training set (70%) and the test set (30%). Each reporter-partner pair was represented in both sets.
- Each model was fit on the training set and then the coefficients were used to predict the test set.

- For each specification, the prediction accuracy was computed using the mean absolute error (MAE) criterion.
- The first three steps were repeated 200 times.

31. As expected, the preferred specification M1.1 better predicts trade on average (Table 5). The second-best model is M1.2, the third M1.3, then M1.4 and finally M1.5, as can also attest the decreasing McFadden pseudo-R squared between model M1.1 and M1.5 (see Annex C for exports and Annex D for imports).

Table 5. Gravity model specifications

		Full model (M1.1)	Reduced model (M1.2)	Reduced model (M1.3)	Reduced model (M1.4)	Reduced model (M1.5)
Old BaTIS specification	Distance	X	X	X	X	X
Old BaTIS specification	Contiguity	X	X	X	X	X
Old BaTIS specification	Common language	X	X	X	X	X
Old BaTIS specification	Colony	X	X	X	X	X
Old BaTIS specification	GDP reporter	X	X	X	X	X
Old BaTIS specification	GDP partner	X	X	X	X	X
Old BaTIS specification	GDP per capita (reporter)	X	X	X	X	X
Old BaTIS specification	Merchandise trade	X	X			
Old BaTIS specification	Tourist arrivals/departures	X		X		
Old BaTIS specification	Time trend (linear)	X	X	X	X	X
Old BaTIS specification	Partner FE	X	X	X	X	
Additional variable	Trade in services with world	X	X	X	X	X

Source: OECD-WTO (2023), Balanced Trade in Services (BaTIS) database.

32. The inclusion of reporter, partner and trading pair fixed effects is recommended in the empirical literature to capture omitted variables and to account for unobservable 'multilateral resistance' terms. The multilateral resistance terms capture the fact that a country's trade depends on the trade frictions across all possible destinations, not just on bilateral trade costs for a specific country pair. (Anderson and Van Wincoop, 2003). However, a theory-consistent set of fixed effects cannot be used in BaTIS due to the need for out-of-sample predictions. Consequently, the reporter's GDP per capita was used as a proxy for (some) of the unobservable characteristics of the reporter, a linear time trend was added to capture global developments in trade, and partner fixed effects were used, where possible, to absorb any omitted variable correlated with a specific trading partner. The model selection builds on the work carried out for the first edition of BaTIS (Fortanier et al., 2017).

33. The models were fitted on the dataset resulting from all the previous estimation steps, using the Poisson Pseudo-Maximum Likelihood estimator (PPML). This method is considered superior to a log-linearised Ordinary Least Squares model (OLS), as it avoids biases in the parameter estimates in the presence of heteroscedasticity and allows for the presence of zero trade flows (Santos Silva and Tenreyro, 2006).

Estimation strategy

34. As the matrix of regressors is incomplete, reduced models had to be used sequentially to derive predictions of total services flows, starting from the full, preferred model described above. Table 6 summarises the specifications of the models together with the percentage of observations and trade value predicted by each model. As the regressions were separately fitted on exports and imports, a total of ten models was used to fill in the gaps in the total services bilateral services exports and imports. The model displaying the lower prediction errors was used for each country pair, on the condition that it predicted at least five data points in the time series. Any gaps resulting from missing predictors were filled in with the backcasting/nowcasting techniques described in step B.1.

Table 6. Summary of the model specifications for the prediction of missing total services, 2005-23

Percentage

		Full model (M1.1)	Reduced model (M1.2)	Reduced model (M1.3)	Reduced model (M1.4)	Reduced model (M1.5)
Distance		X	X	X	X	X
Contiguity		X	X	X	X	X
Common language		X	X	X	X	X
Colony		X	X	X	X	X
GDP reporter		X	X	X	X	X
GDP partner		X	X	X	X	X
GDP per capita (reporter)		X	X	X	X	X
Trade in services with world		X	X	X	X	X
Merchandise trade (bilateral)		X	X			
Tourist arrivals/departures		X		X		
Time trend (linear)		X	X	X	X	X
Partner FE		X	X	X	X	
Exports	**% of total estimated observations**	43.4	3.6	31.9	7.0	14.1
	% of total estimated value	69.5	0.5	26.2	0.6	3.2
Imports	**% of total estimated observations**	24.9	22.9	11.7	26.7	13.8
	% of total estimated value	60.2	8.3	17.9	10.4	3.2

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

35. All missing bilateral flows at the level of individual one-digit or two-digit service categories were estimated using a gravity model for each service category – trade flow combination, with the preferred, full model being complemented by a set of reduced models to deal with gaps in the availability of the explanatory variables.

36. Table 7 summarises the specifications of the models used to predict missing flows for the individual service items, as well as the percentage of data points and trade values predicted by each model.

Table 7. Summary of the model specifications for the prediction of missing individual service categories, 2005-23

Percentage

		Full model (M2.0)	Reduced model (M2.1)	Reduced model (M2.2)	Reduced model (M2.3)
GDP reporter		X	X	X	X
GDP partner		X	X	X	X
Distance		X	X	X	X
Contiguity		X	X	X	X
Common language		X	X	X	X
Colony		X	X	X	X
GDP per capita (reporter)		X	X	X	X
Trade in services with world (relevant category)		X	X	X	X
Time trend (linear)		X	X	X	X
Merchandise trade (bilateral)		X	X		
Partner FE		X	X	X	
Tourist arrivals/departures (travel only)		X			
Exports: one-digit services	**% of total estimated observations**	4.7	52.1	43.2	Not used
	% of total estimated value	17.3	53.1	29.6	Not used
Imports: one-digit services	**% of total estimated observations**	2.8	52.7	44.5	Not used
	% of total estimated value	13.1	55.5	31.4	Not used
Exports: two-digit services	**% of total estimated observations**	8.2	45.4	37.7	8.7
	% of total estimated value	23.7	53.1	22.5	0.7
Imports: two-digit services	**% of total estimated observations**	5.0	39.3	28.0	27.8
	% of total estimated value	20.5	56.4	21.1	1.9

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

37. It is worth emphasising that the a priori order of preference of the models generally provides the most accurate predictions, and that the choice of the model was solely driven by data availability in the regressor matrix for most cases. However, in some specific instances, expert judgement was used to inform the choice of the most suitable model, both for total trade in services and for the individual categories. For instance, when merchandise flows are too concentrated in primary commodities (e.g. oil) and thus very erratic, the merchandise trade variable was deliberately excluded from the explanatory variables. Similarly, partner fixed effects were intentionally excluded when based on very few, quite specific bilateral relationships.

3.3. Towards a complete matrix: Summary of the estimation steps

38. The process described in Sections 2. and 3. resulted in a fully consistent dataset of exports and imports by service category for 202 reporters and partners, from 2005 to 2023. Table 8 provides a summary overview of all the different types of estimates that were produced, showing that more than 90% of the data points and more than half of the trade value in the final bilateral database had to be estimated.

Table 8. Building BaTIS: Reported and estimated flows in the dataset, 2005-23

Percentage

	Total Services		one-digit EBOPS categories		two-digit EBOPS categories	
	% Value	% Count	% Value	% Count	% Value	% Count
Total reported flows	**42**	**10**	**49**	**4**	**33**	**2**
Total estimated flows	**58**	**90**	**51**	**96**	**67**	**98**
Derivations, corrections	20	2	6	22	0	23
Interpolations, back-casting, nowcasting	16	20	14	9	20	7
Gravity estimations	22	68	31	66	47	67
Grand Total	**100**	**100**	**100**	**100**	**100**	**100**

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

3.4. Step B.3: Ensuring the internal consistency of the dataset

39. Having completed all the bilateral trade relationships in the BaTIS matrix, the following step is aimed at ensuring that the dataset is internally consistent in all directions, i.e. that the sum of trade across all partner countries equals the reported values with world as a whole for every children service category, and that the sum across the children service categories equals to total trade in the parent service for each trading pair. To achieve this, firstly the bilateral exports and imports (by service item) were rescaled to match each country's reported trade with the world. Secondly, a bi-proportional adjustment procedure (RAS) was carried out to eliminate any residual discrepancies first between total services and the sum of the 12 one-digit service categories, then between the RAS-adjusted one-digit services and the sum of their two-digit children.

Rescaling of estimates to officially reported world totals

40. For those countries that did not report any geographical detail, the rescaling involved a relatively straightforward proportional scaling of the model-based estimates to the reported world totals (by service category). On the other hand, reported values and higher quality estimates (derivations, back- and nowcasting) were in principle kept as such (not rescaled) if relatively well aligned with the world totals. However, reported values were also rescaled if, despite reporting a nearly complete set of partners, large unallocated trade still existed or if the difference between the sum of (reported) bilateral trade was larger than the world totals (i.e. unallocated trade was negative).

41. Three groups of reporters were therefore identified for which different rescaling strategies were applied. The attribution of countries to groups 2 and 3 was determined based on the ratio between unspecified trade and the sum of the unscaled model estimates. If that ratio was smaller than three, then reporters were allocated to group 3 and only the estimates were rescaled, not the reported data. If the ratio was equal or larger than three, implying a substantial misalignment between the reported bilateral flows and the world totals, the reporter was allocated to group 2 and the reported data was rescaled together with the estimated figures. In summary, the following rescaling rules were applied:

- ***Group 1. Countries that do not report any geographical detail and for which all bilateral flows are derived from gravity-model estimates***

  The gravity-based bilateral estimates were rescaled to match the world totals (by service category). The relative importance of each partner remained the same before and after the rescaling.

- ***Group 2. Countries that report many geographical partners, but their non-geographically specified trade is substantial or negative***

  Both reported data and estimated flows are rescaled to match the world totals (by service category).
- ***Group 3. Countries where data consist of a mix of reported data and model-based estimates, and the amount of unspecified trade is similar to the total amount of trade estimated by the model for the missing partners***

  Reported data (including high quality estimates) were kept as such; model-based estimates were rescaled to benchmark the world totals.

42. In all cases specified above, if no trade with economies outside the BaTIS coverage was reported, 0.5% of total trade was allocated to the partner 'rest of the world' (ROW) to cover any territories beyond the BaTIS scope (see also Section 2.2).

Final bi-proportional scaling (RAS)

43. Any remaining discrepancies firstly between the total services values and the sum of the main one-digit services items, then between the RAS-adjusted one-digit service item and its children, were reconciled using a bi-proportional scaling routine (RAS), which used the reported world totals as constraints to ensure a fully consistent BaTIS matrix.

4. Step C. Balancing of export and import flows

44. Having a complete and consistent BaTIS dataset, the final step involved the reconciliation of the trade asymmetries to produce the final balanced matrix.

45. The balancing process took place in three stages: (i) the asymmetries for exports and imports of total services were reconciled, to arrive at a balanced matrix of total trade in services; (ii) the balancing at the level of the main one-digit service categories (iii) the balancing at the level of the two-digit service categories, were carried out at the level of shares to ensure complete internal consistency in the matrix:

- First, the share of each individual one-digit service category in the (unbalanced) total services flows was calculated; then, those shares were balanced at the bilateral level; and finally, the balanced shares were applied to the balanced value for total services.
- Secondly, the share of each individual two-digit service category in the (unbalanced) parent service flow was calculated; then, those shares were balanced for each two-digit service at the bilateral level; and finally, the balanced shares were applied to the balanced value for the main one-digit parent service.

46. Again, this top-down approach was taken to ensure consistency, but also to recognise that the bilateral data at the higher level of aggregation (totals) are assumed to be of higher quality than those at the level of individual items.

47. The reconciliation procedure entailed the calculation of a weighted average of the reported trade and its mirror flow (i.e. as reported by the partner country), and therefore the balanced value is always between the two flows, at least for total services.[10]

48. The computation of the balancing weights follows the approach developed in cooperation with the Eurostat Full International and Global Accounts for Research in Input-Output (FIGARO) team.

49. First, a bilateral asymmetry indicator was calculated as:

$$A_{ijkt} = \frac{|X_{ijkt} - M_{jikt}|}{X_{ijkt} + M_{jikt}}$$

50. where X and M represent exports and mirror imports, respectively, and I refers to the exporter, j to the importer, k to the service category and t to the year.

51. Second, an asymmetry index was calculated for each exporter (θ_ikt), importer (φ_jkt), service category k and year t as the trade-weighted average of the bilateral asymmetry indicator:

$$\theta_{ikt} = \sum_{j} A_{ijkt} \frac{X_{ijkt} + M_{jikt}}{\sum_j (X_{ijkt} + M_{jikt})}$$

$$\varphi_{jkt} = \sum_{i} A_{ijkt} \frac{X_{ijkt} + M_{jikt}}{\sum_i (X_{ijkt} + M_{jikt})}$$

52. Finally, the complement to 1 of the asymmetry index (symmetry index) was calculated, the three-year moving average of which was used as weight to derive the balanced trade value.

$$SI_{ikt} = (\overline{1 - \theta_{ikt}})$$

$$SI_{jkt} = (\overline{1 - \varphi_{jkt}})$$

53. This method for the calculation of the symmetry indices does not need a discretionary threshold to determine the percentage of ‘good’ bilateral trade flows. Moreover, the use of three-year moving averages further limits any unwanted fluctuations in the symmetry indices. Such stability is important in the process of balancing trade statistics as it avoids introducing potential disruptions in time series solely due to strong variance in the balancing weights.

54. In addition to weighing the bilateral flows by the symmetry index of each partner, the balancing procedure also considers the varying levels of confidence surrounding the different estimation methodologies. For this purpose, four additional weights (w) were applied:

- weight of 1 for reported data or very high-quality estimates (such as derivations from reported items)
- weight of 0.75 for estimates based on information from existing time series (backcasting, nowcasting and interpolations)
- weight of 0.5 for gravity model estimates
- weight of 0.25 for reported data that are considered implausible or incorrect.[11]

55. The final balancing formula between reporter i and partner j, for service item k in year t is therefore as follows:

$$T_{ijkt} = \frac{(SI_{ikt} * w) * X_{ijkt} + (SI_{jkt} * w) * M_{jikt}}{(SI_{ikt} * w) + (SI_{jkt} * w)}$$

56. Table 9 displays the average symmetry index for total services exports and imports (2005-23), for a selection of large service traders, as well as the variation of this symmetry index over time (measured by its standard deviation). Overall, the symmetry indices of large service exporting economies are relatively high and show low standard deviations, while non-OECD countries typically display lower symmetry indices.

Table 9. Symmetry indices for total services trade, 2005-23, selected economies (that report bilateral data)

Index summarised as mean and standard deviation across all years

	Exports		Imports			Exports		Imports	
	mean	sd	mean	sd		mean	sd	mean	sd
Australia	0.82	0.04	0.67	0.07	**Korea**	0.74	0.04	0.76	0.02
Belgium	0.75	0.03	0.83	0.02	**Luxembourg**	0.63	0.04	0.77	0.02
Brazil	0.65	0.03	0.66	0.04	**Malaysia**	0.63	0.03	0.71	0.04
Canada	0.79	0.02	0.87	0.02	**Netherlands**	0.74	0.03	0.75	0.04
China (People's Republic of)	0.73	0.04	0.73	0.03	**New Zealand**	0.75	0.07	0.76	0.02
Denmark	0.70	0.03	0.80	0.03	**Russian Federation**	0.60	0.03	0.68	0.03
France	0.81	0.01	0.85	0.01	**Singapore**	0.65	0.03	0.68	0.05
Germany	0.81	0.01	0.85	0.02	**Spain**	0.83	0.02	0.80	0.02
Hongkong (China)	0.62	0.09	0.64	0.13	**Sweden**	0.74	0.01	0.81	0.01
Ireland	0.74	0.03	0.73	0.05	**Switzerland**	0.79	0.04	0.67	0.02
Italy	0.80	0.02	0.85	0.02	**United Kingdom**	0.80	0.02	0.71	0.03
Japan	0.80	0.02	0.80	0.03	**United States**	0.81	0.03	0.76	0.03

Note: The symmetry index is a measure of the similarity between the reported trade values and the figures reported by their peers; the closer the index is to 1, the more a country's trade is aligned with what is reported by its partners.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

5. Results

5.1. Reported vs. balanced trade for total trade in services

57. The BaTIS balanced trade values, resulting from the reconciliation method described above, differ by construction from both the reported flows and the relevant mirror flows. The size of the difference between the reported and the balanced value is a function of the asymmetry level between two countries as well as the symmetry index of each partner country. In general, the balanced value for a specific service category will be closer to that reported by the country whose statistics are, on average, more in line with what has been reported by its partner country (i.e. the country with a higher symmetry index).

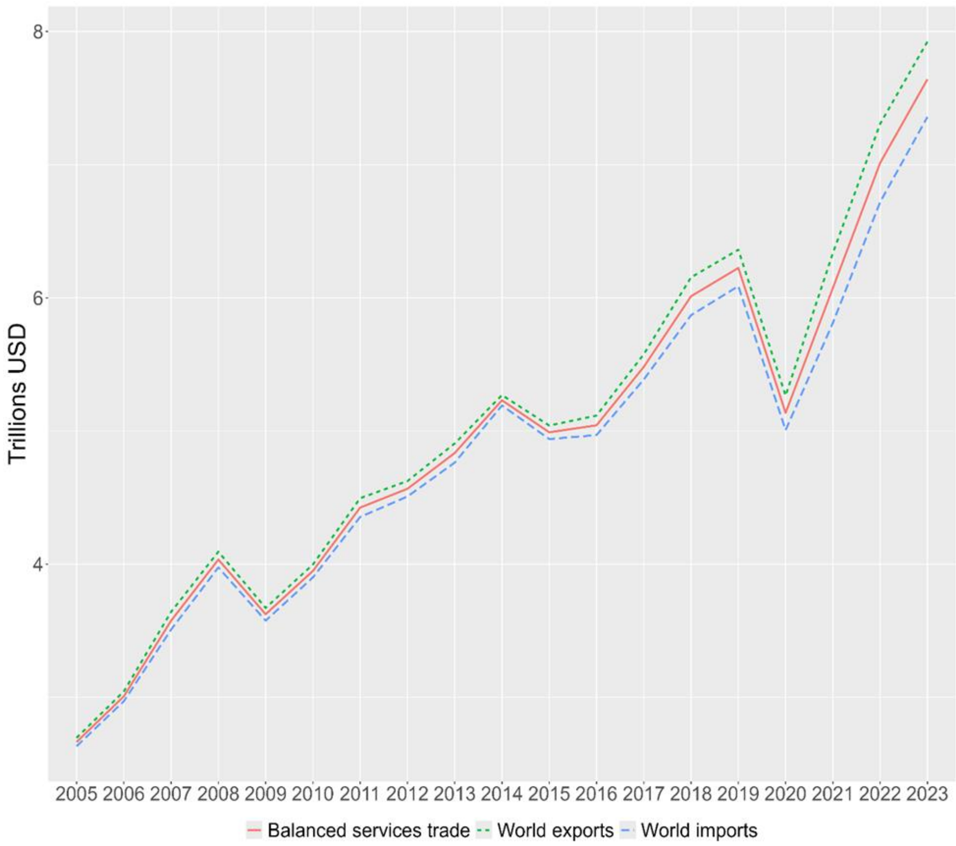
58. According to the BaTIS dataset, world trade in services is estimated at around 7.6 trillion USD in 2023, with Transport, Travel and Other business services, accounting for more than 60% of world services trade. Over the 2005-23 period, services trade has increased by an average annual rate of 6%, overcoming significant troughs during the Great recession (2009) and the Covid-19 pandemic.

59. Digitalisation has had a considerable impact on trade in services, as technological advances allow the remote delivery of many types of services. Hence, over the 2005-23 period, at the world level, digitally deliverable services grew at an average annual rate of 7.4%, higher than the 4.7% recorded for non-digitally deliverable services.

60. Figure 1 shows the comparison between the reported exports and imports and the final balanced trade at world level for total services. Because of the top-down approach followed, BaTIS does not change the overall picture of world trade. However, exports are systematically higher than imports and the gap has increased over time, from about 2% in 2005 to above 8% in 2021-23, in parallel with the growth in overall services trade.

Figure 1. World services trade

Trillions of USD



Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

61. While the difference between reported and balanced values is minimal at world level, it can be quite substantial for individual traders. In particular, it appears that some countries systematically under- or over-report their trade in services compared with what their partners declare, resulting in significant differences between the sum of the balanced trade figures and the original reported totals.

62. Table 10 provides insights into the size of such differences by showing the ratio between the sum of balanced trade and the sum of reported trade across all partners and years for a selection of leading economies that reported bilateral data.

Table 10. Ratio between the sum of balanced bilateral trade and reported total trade across all years, selected countries reporting bilateral trade statistics

Percentage

	Exports		Imports			Exports		Imports	
Australia	6.5	(+)	34.6	(+)	Korea	9.4	(-)	12.0	(-)
Belgium	16.6	(-)	7.3	(-)	Luxembourg	20.2	(-)	1.5	(-)
Brazil	18.4	(+)	3.1	(-)	Malaysia	15.6	(-)	8.6	(-)
Canada	6.9	(-)	2.7	(+)	Netherlands	3.4	(-)	1.2	(+)
China (People's Republic of)	7.0	(-)	22.4	(-)	New Zealand	0.0		13.3	(+)
Denmark	16.5	(-)	10.3	(-)	Russian Federation	11.8	(-)	11.8	(-)
France	9.5	(-)	5.0	(-)	Singapore	3.0	(-)	3.7	(-)
Germany	0.5	(+)	3.0	(+)	Spain	2.8	(-)	17.2	(+)
Hongkong, China	34.6	(+)	45.1	(+)	Sweden	7.0	(-)	2.9	(+)
Ireland	10.1	(-)	8.1	(-)	Switzerland	18.3	(+)	44.8	(+)
Italy	7.1	(+)	1.4	(+)	United Kingdom	1.2	(-)	34.5	(+)
Japan	5.2	(-)	6.1	(-)	United States	9.5	(+)	20.5	(+)

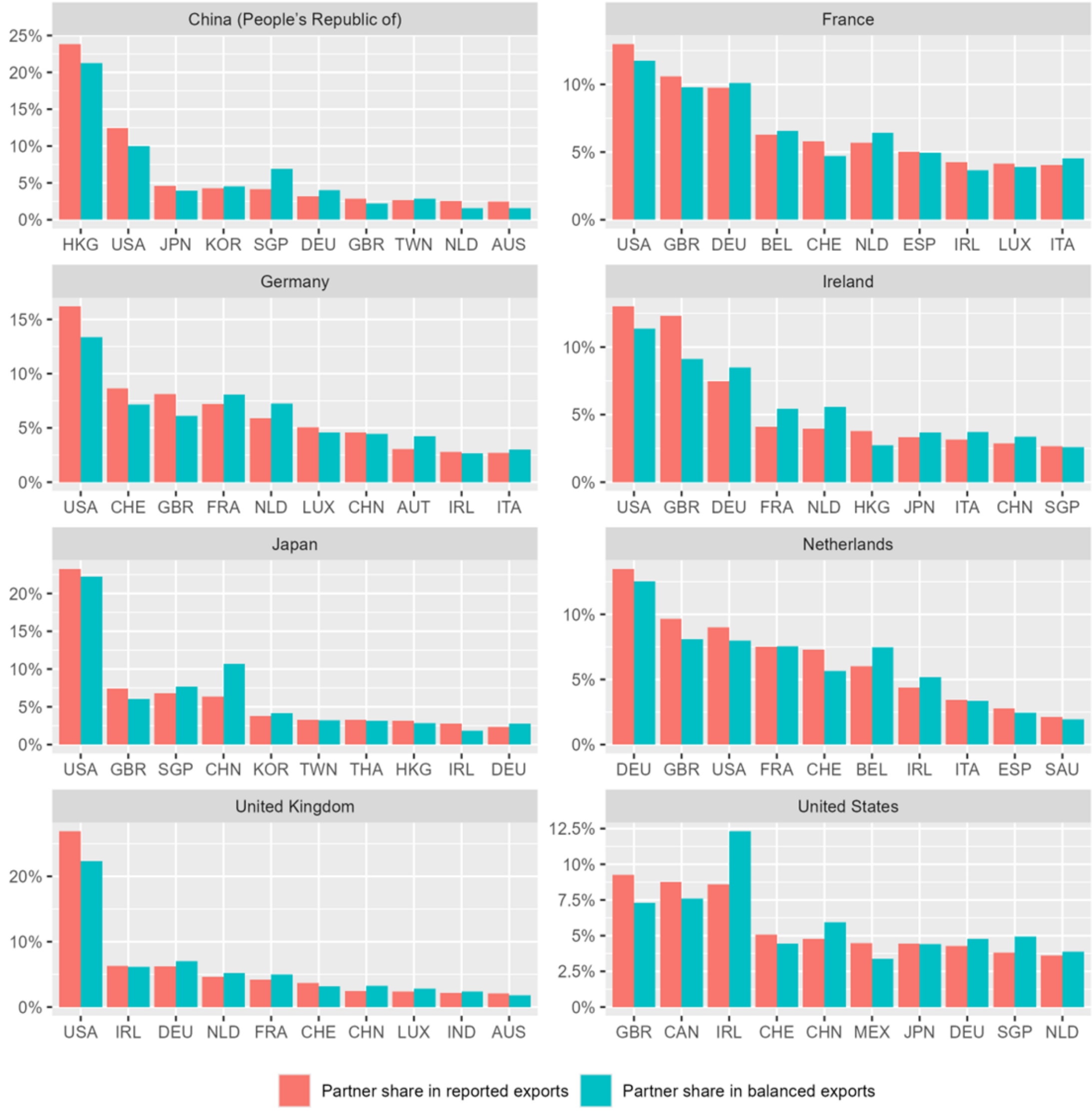
Note: (+) and (-) denote the respectively overall positive or negative difference during the 2005-23 period, with (+) and (-) indicating that balanced trade values were on average higher (lower) than reported statistics.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

63. For some of the leading traders, especially in Europe, the average asymmetries at total services level are relatively low but they remain an issue. For instance, relative intra-EU asymmetries in trade in services showed a median of 10% for total services of EU-27 (see Eurostat, 2023). For Ireland and Luxembourg, for example, both exports and imports appear over-reported, with the difference reaching 20% in the case of Luxembourg's exports. Similarly, China and Singapore seem to over-estimate the level of their exports and imports compared to what their trading partners assess, while the opposite is true for Hong Kong (China). On the other hand, the reported exports of the United Kingdom and the United States are well aligned with the balanced figures, while they both seem to underestimate the level of their imports (by 20% for the United States and 34.5% for the United Kingdom). For Switzerland, which only recently started publishing bilateral trade in services statistics, the balanced figures are significantly higher than reported, particularly on the imports side and for services such as transport, travel and other business services.

64. With a few exceptions, the balancing has a limited impact on the relative importance of the leading traders in world trade and on the shares of the top trading partners for these traders. Figure 2 shows, for a selection of reporting countries with significant trade in services, the share of their top ten partners according to their reported and balanced figures for total services.

Figure 2. Top ten partner countries for selected service exporters: Reported vs. balanced trade data, 2023

Partner share in total exports



Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

65. While for (most of) the OECD economies, the asymmetries are relatively moderate, this is not always the case for other countries covered in BaTIS, including those for which data were imputed or estimated. In some cases, asymmetries can be so large that the total balanced trade figures end up being very different from the reported values.

66. An overview of such extreme cases is provided in Table 11. Similarly to what was already observed in the previous update of BaTIS, Bermuda is at the top of the list: its balanced exports and imports are respectively 44 and 17 times higher than reported over the time period considered. While these figures might seem exaggerated, they reflect the fact that, amongst others, the United States, the Netherlands, Ireland and the United Kingdom report sizeable transactions with Bermuda, most notably in insurance and financial services, which appear to go completely undetected in Bermuda's own data. These issues are

most probably linked to the presence of foreign-controlled MNEs and also occur, although at a smaller scale, in countries such as the Cayman Islands and Barbados.

67. Furthermore, Mexico, Cyprus and the United Arab Emirates all see their balanced exports considerably exceed the reported ones, while the opposite occurs for, amongst others, Macau (China), India and Israel. On the import side, some large non-OECD member economies such as India, Saudi Arabia and Thailand, appear to over-report their trade in services.

Table 11. Extreme cases of under- or over-reporting, total services, selected economies

Percentage, millions of USD

Exports			Imports		
	Ratio between balanced and reported	Average balanced trade value		Ratio between balanced and reported	Average balanced trade value
	2005-23	2005-23		2005-23	2005-23
Bermuda	4 449%	64 165	Bermuda	1 702%	17 275
Liberia	1 311%	2 849	Cayman Islands	670%	8 745
Barbados	669%	10 163	Barbados	618%	4 092
Congo	418%	2 248	Macau (China)	146%	8 845
Afghanistan	201%	2 949	Afghanistan	122%	2 678
Cayman Islands	72%	4 539	Bahamas	89%	3 007
Mauritius	62%	4 151	Liberia	84%	1 115
Bahamas	52%	4 476	Panama	79%	7 155
Cyprus	40%	18 506	Brunei Darussalam	75%	2 776
Nigeria	39%	4 569	Mauritius	75%	3 333
Brunei Darussalam	37%	759	Cyprus	54%	12 632
Tunisia	34%	7 408	Switzerland	45%	169 425
Panama	26%	12 445	Congo	42%	3 875
Mexico	18%	36 499	Mexico	14%	49 061
Switzerland	10%	130 780	Tunisia	6%	3 154
United Arab Emirates	4%	56 074			
			Jordan	-4%	4 172
Sri Lanka	-5%	3 975	Israel	-5%	24 178
Saudi Arabia	-8%	15 338	United Arab Emirates	-5%	59 632
Uzbekistan	-11%	2 003	Uzbekistan	-8%	3 440
Iraq	-15%	3 568	Cambodia	-17%	1 552
Thailand	-16%	38 466	Nigeria	-20%	16 447
Jordan	-25%	4 275	Thailand	-21%	36 445
Israel	-27%	30 529	Qatar	-22%	18 826
Qatar	-29%	9 179	Sri Lanka	-22%	3 647
India	-33%	110 648	Saudi Arabia	-23%	57 378
Macau (China)	-39%	16 547	India	-34%	91 756
Cambodia	-40%	1 761	Kuwait	-34%	14 116
Kuwait	-43%	4 734	Lebanon	-40%	6 622
Lebanon	-49%	6 823	Iraq	-42%	8 194
Iran	-49%	4 485	Turkmenistan	-43%	2 432
Turkmenistan	-62%	1 146	Iran	-49%	8 048

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

5.2. Reported vs. balanced trade for individual service categories

68. As shown in Figure 1, the aggregate absolute differences between the reported data and the final balanced estimates are around 1 to 4% at world level. Those differences are generally much larger, however, for the individual service categories.

69. Table 12 summarises the (trade weighed) average differences between reported and balanced figures, again calculated as the ratio between the sum of the balanced trade and the sum of reported trade across all reporters, partners and years. The ratios are displayed for all the bilateral flows (including the BaTIS estimations) and for the reported flows only.

70. With an absolute ratio of about 3% for both exports and imports, travel is the most aligned category at world level. Transport, instead, appears to be systematically over-reported on the import side. Somewhat surprisingly, reported and balanced values for charges for the use of intellectual property n.i.e seem to be relatively well aligned, possibly reflecting the fact that those services are mostly traded by economies with advanced statistical systems.

71. Two BPM6 items, manufacturing services on physical inputs owned by others and maintenance and repair services n.i.e., seem to be systematically under-reported on the import side (indeed, these items are not yet compiled by many reporters, including a few OECD members). Consequently, balanced trade values in these categories are very close to the reported exports, as the most important exporters in these categories are more likely to report the related transactions.

72. Imports of insurance services are systematically higher than exports, while the opposite is true for financial services, possibly hinting at a misclassification of transactions in this area.

73. High asymmetries are also observable for telecommunication, computer and information services and government goods and services n.i.e, while the discrepancies for construction, personal, cultural and recreational services and, most notably, other business services are relatively moderate at the aggregate level.

Table 12. Ratio between the sum of balanced bilateral trade and reported trade by EBOPS category, across all years

Percentage

	Exports		Imports	
	All observations	Reported flows	All observations	Reported flows
Total services	-2.1	0.0	1.7	8.1
Manufacturing services on physical inputs owned by others	6.5	-15.7	35.3	4.4
Maintenance and repair services n.i.e.	-3.2	-4.9	47.5	15.4
Transport	5.1	3.4	-11.5	2.1
Sea transport	8.6	9.8	-15.1	6.3
Air transport	2.2	-7.8	-9.5	-2.9
Other modes of transport	2.5	-4.4	-8.9	-3.3
Postal and courier services	3.1	-15.2	25.2	6.7
Travel	-2.6	2.8	2.7	4.7
Travel; Business	7.0	-1.7	-0.2	0.6
Travel; Personal	-4.4	0.7	3.4	5.3
Construction	2.4	-3.7	13.8	3.8
Insurance and pension services	26.4	2.6	-21.6	0.4
Financial services	-21.0	-22.1	54.9	49.5
Charges for the use of intellectual property n.i.e.	10.5	5.7	-5.1	-5.7
Telecommunications, computer, and information services	-22.4	-15.4	26.0	25.9
Telecommunications services	-8.0	-2.6	5.8	3.1
Computer services	-26.8	-11.7	32.4	29.7
Information services	-7.8	-19.2	34.8	33.9
Other business services	-1.0	2.6	-2.3	0.3
Research and development services	11.0	13.2	-10.8	-6.3
Professional and management consulting services	-7.5	-2.0	8.5	10.0
Technical, trade-related, and other business services	0.7	8.8	-6.4	8.2
Personal, cultural, and recreational services	2.8	-1.8	5.5	13.1
Audiovisual and related services	11.3	-4.9	9.8	16.0
Personal, cultural, and recreational services other than audiovisual and related services	-8.7	4.5	-0.7	13.3
Government goods and services n.i.e.	29.8	35.6	-1.6	2.8

Note: A positive (negative) ratio indicates that the balanced values are higher (lower) than the reported figures.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

6. Conclusions and next steps

74. This paper presented the OECD-WTO BaTIS dataset, describing the methodology developed by the OECD and WTO to build a complete, consistent and balanced matrix of trade in services statistics.

75. Further work is necessary to improve future editions of the dataset. Over the past years, several countries have been actively involved in bilateral and multilateral efforts to examine the causes of the observed asymmetries, including by sharing compilation methodologies and detailed information.[12] Such efforts at the level of national compilers are crucial to improve the reliability of trade in services statistics at source and to enhance public trust in these data. National efforts to reconcile the observed asymmetries are preferable to any mechanical balancing procedure, and therefore new results made available will be absorbed in the future editions of BaTIS.

76. While the development of BaTIS was driven by the need for complete and coherent trade in services statistics for the purposes of constructing global Supply and Use and Input-Output tables, the dataset also constitutes a standalone analytical product. In particular for countries with less developed statistical systems, BaTIS can serve as a starting point to identify the most important trends and to support trade in services policies, hopefully fostering continuous improvements in existing data compilation practices.

77. BaTIS is intended to be regularly updated and improved. The most prominent areas for future work include:

- Applying the required changes to BaTIS according to the forthcoming seventh edition of the IMF Balance of Payments and International Investment Position Manual (BPM7); and the EBOPS 2026 classification.
- Adding the Mode of Supply dimension. The methodology underpinning the construction of BaTIS is fully compatible with the WTO dataset on Trade in Services by Mode of Supply (TISMOS). Indeed, the two products were developed with a view to be eventually merged. A detailed, multidimensional dataset providing information on the type of service traded, the partner and the mode of supply, whereas challenging to construct, would constitute an invaluable tool for policy-makers and trade negotiators. Furthermore, given the inherent relationship between the supply of services via mode one (cross border supply) and the digital delivery of services (see IMF et al., 2023), the addition of the mode of supply dimension to BaTIS would allow a deeper understanding of how services are traded in the 21st century.

Notes

[1] The first edition of BaTIS followed the fifth edition of the IMF Balance of Payments and International Investment Position Manual (BPM5) and included annual data from 1995 to 2012, covering 191 economies and the 11 main service categories, in line with the Extended Balance of Payments Services Classification (EBOPS) 2002. The second edition of BaTIS, based on BPM6, provided annual data from 2005 to 2019, covering 202 economies and the 12 main EBOPS 2010 service categories. The third edition of BaTIS was an update of the second edition, covering the 2005-21 period.

[2] Available at https://stats.wto.org/.

[3] In addition to total services and the 26 service categories specified in EBOPS 2010, Table 1 presents a few memorandum items derived from the standard components and available in BaTIS. Commercial services, in particular, is used in the WTO context and excludes government goods and services n.i.e. because "services supplied in the exercise of governmental authority", that is neither on a commercial basis nor in competition with other suppliers, are excluded from the General Agreement on Trade in Services (GATS).

[4] With the changeover to the BPM6 standards, a smoothing method was introduced to limit the occurrence of negative flows and to facilitate the economic interpretation of such flows, as well as better alignment with SNA aggregates, and, in particular with the Supply and Use tables. Indeed, negative flows are much less common in BPM6 than in BPM5, but they can still occur.

[5] Complementary national data sources include information published by institutions other than the main provider of balance of payments data. Those can cover a subset of services (most commonly travel) and sometimes are not entirely consistent with the official balance of payments statistics.

[6] The corrections made to insurance and pension services, manufacturing services on physical inputs owned by others, and maintenance and repair services were given the lowest weight in the final balancing procedure (see Section 4.).

[7] The CEPII GeoDist database is available at www.cepii.fr/cepii/en/bdd_modele/bdd.asp. The manual imputations can be provided upon request.

[8] https://unctadstat.unctad.org.

[9] UNWTO Compendium of Tourism Statistics dataset, www.unwto.org/tourism-statistics/tourism-statistics-database.

[10] The reconciled values for the individual service items, instead, may be outside the range of the reported and mirror value as the balancing was performed on the shares, not on the values themselves.

[11] These cases include zero trade reported for items that are in fact not compiled (may happen for manufacturing services on inputs owned by others, for instance), or corrections of reported negative insurance flows (see Section 2.2).

[12] A number of bilateral and trilateral meetings have been held in the context of the OECD Working Party on Trade in Goods and Services (WPTGS).

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Annex A. List of economies covered in the BaTIS dataset

List of reporters and partners:

Afghanistan, Albania, Algeria, Angola, Anguilla, Antigua and Barbuda, Argentina, Armenia, Aruba, Australia, Austria, Azerbaijan, Bahamas, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Belize, Benin, Bermuda, Bhutan, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Brunei Darussalam, Bulgaria, Burkina Faso, Burundi, Cabo Verde, Cambodia, Cameroon, Canada, Cayman Islands, Central African Republic, Chad, Chile, People's Republic of China, , Colombia, Comoros, Congo, Costa Rica, Côte d'Ivoire, Croatia, Cuba, Curaçao, Cyprus, Czech Republic, Democratic Republic of the Congo, Denmark, Djibouti, Dominica, Dominican Republic, Ecuador, Egypt, El Salvador, Equatorial Guinea, Eritrea, Estonia, Eswatini, Ethiopia, Faeroe Islands, Fiji, Finland, France, French Polynesia, Gabon, Georgia, Germany, Ghana, Greece, Grenada, Guatemala, Guinea, Guinea-Bissau, Guyana, Haiti, Honduras, Hong Kong (China), Hungary, Iceland, India, Indonesia, Islamic Republic of Iran, Iraq, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Kiribati, Democratic People's Republic of Korea, Korea, Kuwait, Kyrgyzstan, Lao People's Democratic Republic, Latvia, Lebanon, Lesotho, Liberia, Libya, Lithuania, Luxembourg, Macau (China), Madagascar, Malawi, Malaysia, Maldives, Mali, Malta, Mauritania, Mauritius, Mexico, Republic of Moldova, Mongolia, Montenegro, Montserrat, Morocco, Mozambique, Myanmar, Namibia, Nepal, Netherlands, Netherlands Antilles, New Caledonia, New Zealand, Nicaragua, Niger, Nigeria, Republic of North Macedonia, Norway, Oman, Pakistan, Palestinian Authority or West Bank and Gaza Strip, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Qatar, Romania, Russian Federation, Rwanda, Saint Kitts and Nevis, Saint Lucia, Saint Martin, Saint Vincent and the Grenadines, Samoa, Sao Tomé and Principe, Saudi Arabia, Senegal, Serbia, Former Serbia and Montenegro, Seychelles, Sierra Leone, Singapore, Slovak Republic, Slovenia, Solomon Islands, Somalia, South Africa, Spain, Sri Lanka, Sudan, Suriname, Sweden, Switzerland, Syrian Arab Republic, Chinese Taipei Tajikistan, United Republic of Tanzania, Thailand, The Gambia, Timor-Leste, Togo, Tonga, Trinidad and Tobago, Tunisia, Republic of Türkiye, Turkmenistan, Turks and Caicos Islands, Tuvalu, Uganda, Ukraine, United Arab Emirates, United Kingdom, United States, *Kosovo*[1], Uruguay, Uzbekistan, Vanuatu, Bolivarian Republic of Venezuela, Viet Nam, Yemen, Zambia, Zimbabwe.

List of partners covered in entity "Rest of the World" (ROW):

Andorra; Antarctica; Samoa; Bonaire; Saint Eustatius; Saba; Bouvet Island; Cocos (Keeling) Islands; Cook Islands; Christmas Island; Western Sahara; Falkland Islands (Malvinas); Federated States of Micronesia; French Southern and Antarctic Lands; French Guiana; Guernsey; Gibraltar; Greenland; Guadeloupe; South Georgia and the South Sandwich Islands; Guam; Heard Island and McDonald Islands; Isle of Man; British Indian Ocean Territory; Jersey; Liechtenstein; Monaco; Marshall Islands; Northern Mariana Islands; Martinique; Norfolk Island; Nauru; Niue; Saint Pierre and Miquelon; Pitcairn; Puerto Rico; Palau; Réunion; Saint Helena, Ascension and Tristan da Cunha; San Marino; South Sudan; Tokelau; United States Minor Outlying Islands; Holy See; British Virgin Islands; United States Virgin Islands; Wallis and Futuna Islands.

[1] The designation is without prejudice to positions on status, and is in line with United Nations Security Council resolution 1244/1999 and the Advisory Opinion of the International Court of Justice on Kosovo's declaration of independence

Annex B. Reported data availability: Maximum number of partners by reporter and EBOPS 2010 category

ISO3	Partner name	S	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SC1	SC2	SC3	SC4	SDA	SDB	SI1	SI2	SI3	SJ1	SJ2	SJ3	SK1	SK2
XKV	*Kosovo*[1]	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
ALB	Albania	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
ARG	Argentina	194	3	65	176	185	10	176	65	59	..	134	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
AUS	Australia	34	34	34	34	34	34	34	34	34	34	34	34	34	..	..	..	..	34	34	..	..	..	..	..	..	..	..
AUT	Austria	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
BGD	Bangladesh	193	81	45	171	189	34	51	93	43	158	164	121	104	164	132	39	29	..	..	75	158	54	46	136	155	..	..
BLR	Belarus	187	41	69	160	161	53	134	157	96	171	128	103	114	113	138	110	75	..	..	131	157	57	49	105	116	78	101
BEL	Belgium	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
BTN	Bhutan	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
BOL	Bolivia	..	..	..	..	28	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
BIH	Bosnia and Herzegovina	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
BRA	Brazil	37	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
BGR	Bulgaria	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
CAN	Canada	58	..	83	58	58	83	83	83	83	83	81	83	41	1	1	1	1	1	1	83	83	1	83	83	81	83	83
CHL	Chile	..	..	..	31	12	..	..	..	..	6	14	..	..	..	..	..	..	..	..	..	6	..	..	..	..	..	..
CHN	China (People's Republic of)	13	13	13	13	13	13	13	13	13	13	13	13	13	..	..	..	..	..	..	..	..	..	..	..	..	..	..
COL	Colombia	..	3	17	42	1	2	..	..	32	57	62	44	1	23	26	34	6	..	..	33	..	53	18	..	27	13	..
CRI	Costa Rica	200	200	200	200	..	200	200	200	200	200	200	200	200	..	..	..	..	..	..	200	..	200	200	200	200	200	200
HRV	Croatia	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
CYP	Cyprus	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59

ISO3	Partner name	S	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SC1	SC2	SC3	SC4	SD A	SD B	SI1	SI2	SI3	SJ1	SJ2	SJ3	SK1	SK2
CZE	Czech Republic	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
DNK	Denmark	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
EGY	Egypt	..	114	119	149	137	100	94	126	135	131	139	107	132	..	18	114	51	..	..	91	..	..	..	..	..	..	..
EST	Estonia	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
FIN	Finland	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
FRA	France	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
DEU	Germany	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
GRC	Greece	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
HKG	Hong Kong (China)	46	1	7	46	21	..	43	42	40	40	44	41	..	23	21	18	15	..	..	19	19	18	18	23	22	16	16
HUN	Hungary	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
ISL	Iceland	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
IRL	Ireland	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
ISR	Israel	..	..	37	..	38	31	..	15	37	28	..	16	..	..	..	..	..	..	..	38	44	29	35	..	9	16	43
ITA	Italy	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
JPN	Japan	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	..	..	..	..	..	..	..	..	..	..	..	..
KAZ	Kazakhstan	..	11	59	117	91	20	..	24	28	67	72	37	59	..	..	..	..	..	..	20	39	32	5	..	21	..	..
KOR	Korea	4	4	..	4	4	..	4	..	4	4	4	..	4	..	..	..	..	..	..	..	..	..	..	..	..	..	..
LVA	Latvia	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
LTU	Lithuania	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
LUX	Luxembourg	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
MYS	Malaysia	47	20	..	20	20	20	..	..	..	20	20	20	20	..	..	..	..	..	..	..	..	..	..	..	..	..	..
MLT	Malta	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
MEX	Mexico	67	20	18	44	3	..	..	..	15	25	39	9	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
MNE	Montenegro	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
NLD	Netherlands	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
NZL	New Zealand	198	..	53	58	194	58	58	58	58	58	58	58	190	7	8	..	6	191	196	12	19	10	15	33	32	5	13
MKD	North Macedonia	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
NOR	Norway	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59

ISO3	Partner name	S	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SC1	SC2	SC3	SC4	SDA	SDB	SI1	SI2	SI3	SJ1	SJ2	SJ3	SK1	SK2
PAK	Pakistan	34	..	68	184	152	49	183	174	88	161	182	64	127	..	..	..	..	..	..	..	..	..	..	..	..	..	..
PER	Peru	..	..	..	..	71	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
POL	Poland	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
PRT	Portugal	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
ROU	Romania	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
RUS	Russian Federation	197	81	129	176	81	100	149	192	115	184	184	136	1	79	79	79	79	..	..	79	79	79	79	79	79	79	..
SRB	Serbia	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
SGP	Singapore	31	9	9	9	..	9	9	9	9	9	9	9	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
SVK	Slovak Republic	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
SVN	Slovenia	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
ESP	Spain	199	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
SWE	Sweden	200	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59
CHE	Switzerland	40	..	..	40	40	..	40	40	40	40	40	..	..	..	..	..	..	..	..	..	..	..	16	40	40	..	..
TTO	Trinidad and Tobago	..	..	..	..	..	3	..	3	3	3	3	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
TUN	Tunisia	65	..	..	131	103	..	..	..	..	..	..	..	70	..	..	..	..	..	..	..	..	..	..	..	..	..	..
TUR	Turkey	38	38	38	38	59	1	38	38	38	38	38	38	1	1	1	1	59	1	1	59	1	59	1	59	1	1	1
UKR	Ukraine	6	6	6	6	6	6	6	6	6	6	6	6	5	..	..	..	..	..	..	..	..	..	..	..	..	..	..
GBR	United Kingdom	200	59	59	59	63	59	59	59	59	60	61	59	59	59	59	59	59	60	61	59	59	59	59	59	59	59	59
USA	United States	73	..	73	73	73	64	73	73	73	73	73	73	60	73	73	73	..	73	73	73	71	73	69	71	71	71	71

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

[1] The designation is without prejudice to positions on status, and is in line with United Nations Security Council resolution 1244/1999 and the Advisory Opinion of the International Court of Justice on Kosovo's declaration of independence.

Annex C. Results of PPML regressions for total services (exports)

	Model 1	Model 2	Model 3	Model 4	Model 5
Constant	-9.10***	-7.53***	7.63***	6.27***	22.64***
	-1.28	-1.11	-1.16	-1.03	-1.13
Distance	-0.21***	-0.21***	-0.43***	-0.44***	-0.53***
	0	0	0	0	0
Contiguity	0.13***	0.15***	0.34***	0.33***	0.20***
	-0.01	-0.01	-0.01	-0.01	-0.01
Common language	0.38***	0.38***	0.50***	0.51***	0.66***
	-0.01	-0.01	-0.01	-0.01	-0.01
Colony	0.16***	0.17***	0.08***	0.09***	0.22***
	-0.01	-0.01	-0.01	-0.01	-0.01
GDP of reporter	-0.04***	-0.05***	0.19***	0.18***	0.20***
	0	0	0	0	0
GDP of partner	0.24***	0.22***	0.44***	0.42***	0.74***
	-0.01	-0.01	-0.01	-0.01	0
GDP/capita of reporter	0.08***	0.07***	-0.09***	-0.07***	-0.11***
	-0.01	-0.01	-0.01	0	-0.01
Total trade with world	0.65***	0.67***	0.83***	0.81***	0.79***
	-0.01	-0.01	-0.01	0	-0.01
Merchandise trade	0.37***	0.37***			
	0	0			
Arrivals	0.02***		-0.02***		
	0		0		
t	0.00**	0.00*	-0.01***	-0.00***	-0.01***
	0	0	0	0	0
Partner FE	Y	Y	Y	Y	Y
Deviance	16 839 122.86	19 911 293.59	23 993 456.94	28 704 368.57	49 986 801.61
Num. obs.	94 687.00	116 035.00	120 373.00	148 981.00	148 981.00
Mc Fadden pseudo R2	0.94	0.94	0.93	0.93	0.87

Note: Significance levels indicated with *** for 0.1%, ** for 1% and * for 5%. The McFadden pseudo- R2 is defined as one minus the ratio between the (maximised) likelihood value from the fitted model, and the corresponding value for the null model.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

Annex D. Results of PPML regressions for total services (imports)

	Model 1	Model 2	Model 3	Model 4	Model 5
Constant	2.98*	-7.75***	14.44***	5.06***	29.09***
	-1.31	-1.15	-1.42	-1.23	-2.29
Distance	-0.21***	-0.21***	-0.43***	-0.42***	-0.54***
	0	0	0	0	-0.01
Contiguity	0.14***	0.15***	0.24***	0.24***	0.05**
	-0.01	-0.01	-0.01	-0.01	-0.02
Common language	0.32***	0.33***	0.51***	0.52***	0.69***
	-0.01	-0.01	-0.01	-0.01	-0.01
Colony	0	-0.01	-0.05***	-0.06***	0.11***
	-0.01	-0.01	-0.01	-0.01	-0.02
GDP of reporter	0.02***	-0.06***	0.24***	0.17***	0.17***
	0	0	0	0	-0.01
GDP of partner	0.16***	0.18***	0.30***	0.32***	0.72***
	-0.02	-0.01	-0.02	-0.01	0
GDP/capita of reporter	-0.04***	-0.01	-0.10***	-0.07***	-0.15***
	0	0	-0.01	0	-0.01
Total trade with world	0.75***	0.73***	0.89***	0.86***	0.89***
	-0.01	0	-0.01	-0.01	-0.01
Merchandise trade	0.40***	0.38***			
	0	0			
Departures	-0.14***		-0.12***		
	0		0		
t	-0.00***	0.00**	-0.01***	-0.01***	-0.02***
	0	0	0	0	0
Partner FE	Y	Y	Y	Y	Y
Deviance	16 985 168.57	20 400 852.34	24 551 368.79	29 313 863.14	55 850 590.42
Num. obs.	87 001	107 307	111 523	140 468	140 468
Mc Fadden pseudo R2	0.93	0.93	0.92	0.92	0.84

Note: Significance levels indicated with *** for 0.1%, ** for 1% and * for 5%. The McFadden pseudo- R2 is defined as one minus the ratio between the (maximised) likelihood value from the fitted model, and the corresponding value for the null model.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

Annex E. Estimated bilateral trade by model, service category and flow

Percentage, across all years

Item	Description	Model	Exports: % of total est. obs.	Exports: % of total est. val.	Imports: % of total est. obs.	Imports: % of total est. val.
SA	Manufacturing services on physical inputs owned by others	M2.1	64	79	64	89
		M2.2	36	21	36	11
SB	Maintenance and repair services n.i.e.	M2.1	60	64	58	74
		M2.2	40	36	42	26
SC	Transport	M2.1	54	74	54	73
		M2.2	46	26	46	27
SC1	Sea transport	M2.1	50	86	53	84
		M2.2	32	13	35	15
		M2.3	18	1	13	1
SC2	Air transport	M2.1	49	71	48	71
		M2.2	36	28	33	28
		M2.3	15	1	19	2
SC3	Other modes of transport	M2.1	52	67	48	59
		M2.2	29	28	26	35
		M2.3	19	5	26	6
SC4	Postal and courier services	M2.1	33	57	35	64
		M2.2	20	21	17	25
		M2.3	47	22	48	11
SD	Travel	M2.0	47	67	30	63
		M2.1	4	1	22	6
		M2.2	49	33	48	31
SDA	Travel; Business	M2.0	49	73	32	71
		M2.1	3	3	22	7
		M2.2	45	24	45	23
		M2.3	3	0	1	0
SDB	Travel; Personal	M2.0	49	71	32	70
		M2.1	4	3	22	6
		M2.2	46	26	45	24
		M2.3	1	0	1	0
SE	Construction	M2.1	59	75	54	47
		M2.2	41	25	46	53
SF	Insurance and pension services	M2.1	57	51	55	46
		M2.2	43	49	45	54
SG	Financial services	M2.1	56	62	55	67
		M2.2	44	38	45	33
SH	Charges for the use of intellectual property n.i.e.	M2.1	61	69	57	70
		M2.2	39	31	43	30
SI	Telecommunications, computer, and information services	M2.1	55	74	55	79

Item	Description	Model	Exports		Imports	
			% of total est. obs.	**% of total est. val.**	**% of total est. obs.**	**% of total est. val.**
		M2.2	45	26	45	21
SI1	Telecommunications services	M2.1	56	59	44	68
		M2.2	44	41	27	30
		M2.3	1	0	29	2
SI2	Computer services	M2.1	55	79	38	78
		M2.2	37	21	23	18
		M2.3	8	0	39	4
SI3	Information services	M2.1	59	75	34	62
		M2.2	40	25	15	27
		M2.3	1	0	51	10
SJ	Other business services	M2.1	55	76	54	74
		M2.2	45	24	46	26
SJ1	Research and development services	M2.1	63	81	37	80
		M2.2	36	19	15	14
		M2.3	1	0	48	6
SJ2	Professional and management consulting services	M2.1	59	86	44	85
		M2.2	41	14	28	13
		M2.3	0	0	27	2
SJ3	Technical, trade-related, and other business services	M2.1	56	77	50	79
		M2.2	43	23	34	20
		M2.3	1	0	16	1
SK	Personal, cultural, and recreational services	M2.1	61	73	58	69
		M2.2	39	27	42	31
SK1	Audiovisual and related services	M2.1	63	83	35	59
		M2.2	37	16	16	24
		M2.3	0	1	49	17
SK2	Personal, cultural, and recreational services other than audiovisual and related services	M2.1	60	71	34	61
		M2.2	39	28	16	17
		M2.3	1	1	50	22
SL	Government goods and services n.i.e.	M2.1	57	46	56	35
		M2.2	43	54	44	65

Note: The percentages refer to the gravity-based estimates only, i.e. they exclude the reported data as well as the other types of estimates.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

Annex F. Assessing the predictive power of the gravity models

This Annex describes the cross-validation exercise carried out to measure the out-of-sample accuracy of the models used to predict total services trade. In particular, the predictive power of the five models used in the previous edition of BaTIS was compared with that of the same models but with the addition of the trade in services with partner world variable for all five specifications.

		Full model (M1.1)	Reduced model (M1.2)	Reduced model (M1.3)	Reduced model (M1.4)	Reduced model (M1.5)
Old BaTIS specification	Distance	X	X	X	X	X
Old BaTIS specification	Contiguity	X	X	X	X	X
Old BaTIS specification	Common language	X	X	X	X	X
Old BaTIS specification	Colony	X	X	X	X	X
Old BaTIS specification	GDP reporter	X	X	X	X	X
Old BaTIS specification	GDP partner	X	X	X	X	X
Old BaTIS specification	GDP per capita (reporter)	X	X	X	X	X
Old BaTIS specification	Merchandise trade	X	X			
Old BaTIS specification	Tourist arrivals/departures	X		X		
Old BaTIS specification	Time trend (linear)	X	X	X	X	X
Old BaTIS specification	Partner FE	X	X	X	X	
Additional variable	Trade in services with world	X	X	X	X	X

Source: OECD-WTO (2023), BaTIS database.

The sample used for the cross-validation contained information available for all specifications, i.e. the sample of model M1.1. The algorithm took the following steps:

i. The sample was randomly split into two datasets: the training set (70%) and the test set (30%). Each reporter-partner pair was represented in both sets.

ii. Each model was fit on the training set and then the coefficients were used to predict the test set.

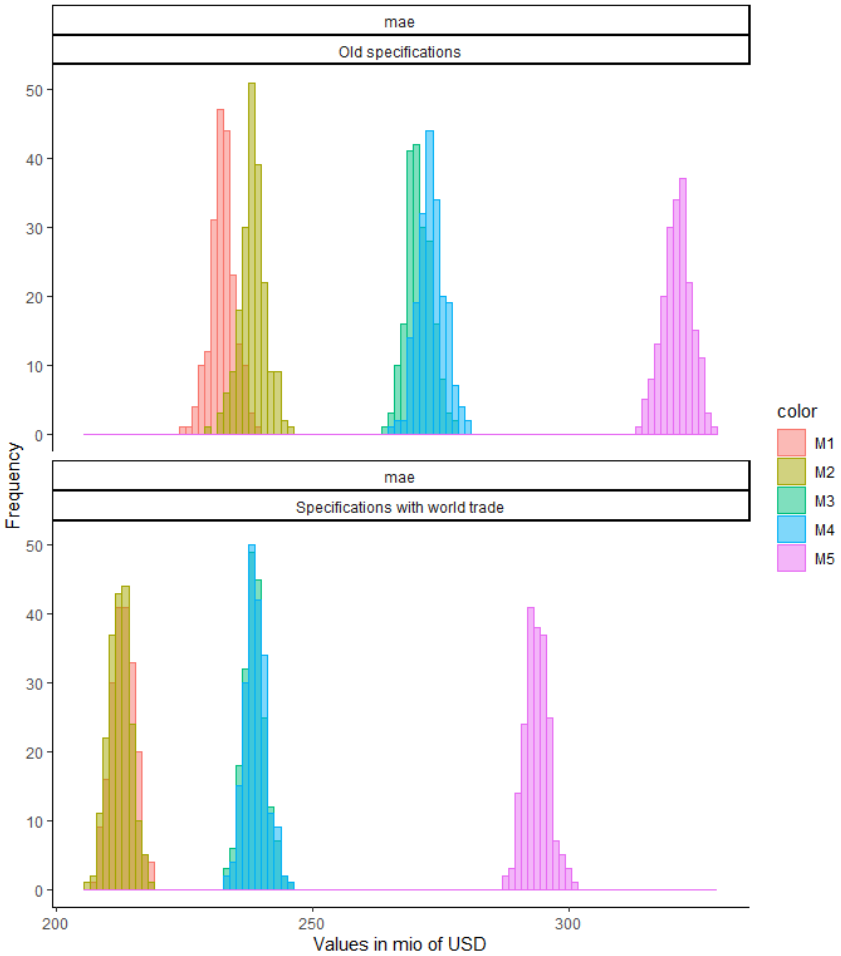
iii. For each specification M1.1, M1.2, M1.3, M1.4, and M1.5, the prediction accuracy was computed using the mean absolute error (MAE) criterion.

iv. Steps i. to iii. were repeated 200 times.

The resulting distributions of the MAE are plotted in Figure A F.1, with the top panel including the old specifications and the bottom panel covering the trade with world as additional predictor. As expected, the preferred specification M1.1 better predicts trade on average. The second-best model is M1.2, the third M1.3, then M1.4 and finally M1.5. This means that, starting from the most parsimonious model (M1.5), adding each explanatory variable provides useful information and enables to better predict bilateral trade. The bottom panel shows that the accuracy of the out of sample predictions improves significantly (for all five specifications) when total trade in services with partner world is added as explanatory variable, despite the obvious endogeneity with the response variable. Although small values appeared more difficult to predict, these results hold when the sample is split between large trade flows (above 50 million USD) and smaller flows (below 50 million of USD). Also, changing the proportions of the training and test sets or the number of repetitions gave a similar picture.

Figure A F.1. Accuracy of predictions for total services trade: Mean absolute error

Million USD



Source: OECD-WTO (2023), Balanced Trade in Services (BaTIS) database.

Annex G. McFadden pseudo R-squared for the gravity estimations of individual service categories, by model and flow

		Exports				Imports			
Service category	**Code**	**M2.0**	**M2.1**	**M2.2**	**M2.3**	**M2.0**	**M2.1**	**M2.2**	**M2.3**
Manufacturing services on physical inputs owned by others	SA		80	80	70		89	89	82
Maintenance and repair services n.i.e.	SB		63	64	53		89	88	83
Transport	SC		89	88	83		89	88	82
Sea transport	SC1		87	87	76		83	81	64
Air transport	SC2		87	85	81		87	86	78
Other modes of transport	SC3		85	84	77		88	86	78
Postal and courier services	SC4		87	86	76		89	88	76
Travel	SD	90	89	89	85	91	91	90	82
Travel; Business	SDA	92	91	89	84	91	90	87	84
Travel; Personal	SDB	91	89	87	83	92	92	90	76
Construction	SE		76	74	66		75	75	68
Insurance and pension services	SF		89	89	80		87	91	63
Financial services	SG		94	94	83		89	90	74
Charges for the use of intellectual property n.i.e.	SH		92	92	80		89	90	75
Telecommunications, computer, and information services	SI		91	91	86		90	90	78
Telecommunications services	SI1		86	85	78		84	82	72
Computer services	SI2		90	90	84		91	89	73
Information services	SI3		88	88	82		86	85	73
Other business services	SJ		94	93	84		93	93	86
Research and development services	SJ1		90	89	69		91	91	83
Professional and management consulting services	SJ2		95	94	82		93	92	83
Technical, trade-related, and other business services	SJ3		91	90	84		90	89	82
Personal, cultural, and recreational services	SK		90	89	85		84	84	77
Audiovisual and related services	SK1		92	92	87		90	88	79
Personal, cultural, and recreational services other than audiovisual and related services	SK2		83	81	75		67	66	54
Government goods and services n.i.e.	SL		83	83	71		64	64	47

Note: The McFadden pseudo R2 is defined as one minus the ratio between the (maximised) likelihood value from the fitted model, and the corresponding value for the null model. The full regression tables at the level of individual service categories are available upon request.
Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.

Annex H. Methodology codes used in BaTIS

Methodology	Description
R_OECD	Reported: OECD International Trade in Services Statistics (ITSS).
R_EU_ITS	Reported: EUROSTAT International Trade in Services statistics.
R_EU_EBOP	Reported: EUROSTAT Quarterly Balance of Payments statistics.
R_NAT	Reported: National source.
R_NAT.1	Reported: National source, but different from main balance of payments provider. Data rescaled to fit world totals.
R_UNSD	Reported: United Nations Statistics Division (Comtrade).
R_WTO_UNCTAD	Reported: WTO UNCTAD (partner world only).
E0	Estimated as negligible/zero.
E1	Simple derivation from reported figures.
E1.2	Simple derivation - after backcasting/nowcasting/interpolation.
E2	Estimation of most recent year(s) missing in primary source by using the national BOP growth rate (partner world only).
E2.1	Estimation of most recent year(s) missing in primary source by using the national BOP growth rate. Past shares used to estimate subitems (partner world only).
E3	Estimated using past or future shares (partner world only).
E4	Correction of mistakes in source data, such as implausible negative values, definition not in line with international recommendations, etc. (partner world only).
E5	Estimates based on regional growth rates (partner world only).
E6	Correction for negative insurance.
E7	Completely missing time series information. Estimated using reported shares within a cluster of similar economies for a given year (partner world only).
E8	Gaps in reported time series: estimated by back/nowcasting or interpolation (partner world only).
E8.0	Gaps in reported time series: estimated as zero using interpolation or back-nowcasting (bilateral data).
E8.1	Gaps in reported time series: estimated as non-zero value using interpolation or back-nowcasting (bilateral data). As only one data point is reported, the same share is applied to all years.
E8.2	Gaps in reported time series: estimated as non-zero value using interpolation or back-nowcasting (bilateral data).
M1.1	Estimation of total services from gravity model: specification (M1.1).
M1.2	Estimation of total services from gravity model: specification (M1.2).
M1.3	Estimation of total services from gravity model: specification (M1.3).
M1.4	Estimation of total services from gravity model: specification (M1.4).
M1.5	Estimation of total services from gravity model: specification (M1.5).
M2.0	Estimation of service item from gravity model: specification (M2.0).
M2.1	Estimation of service item from gravity model: specification (M2.1).
M2.2	Estimation of service item from gravity model: specification (M2.2).
M2.3	Estimation of service item from gravity model: specification (M2.3).
M8.0	Remaining gaps in total services after prediction from gravity models: estimated as zero using interpolation or back-nowcasting on predicted data.
M8.2	Remaining gaps in total services after prediction from gravity models: estimated as non-zero value using interpolation or back-nowcasting on predicted data.
M8.3	Imputation of Rest of the World aggregate.
W0	Derived as zero, as partner world is zero.
A	Aggregation of figures with different methodology codes (used for country groups).
Z	Reporter or partner country not available at the time of analysis.

Source: OECD-WTO (2025), Balanced Trade in Services (BaTIS) database.